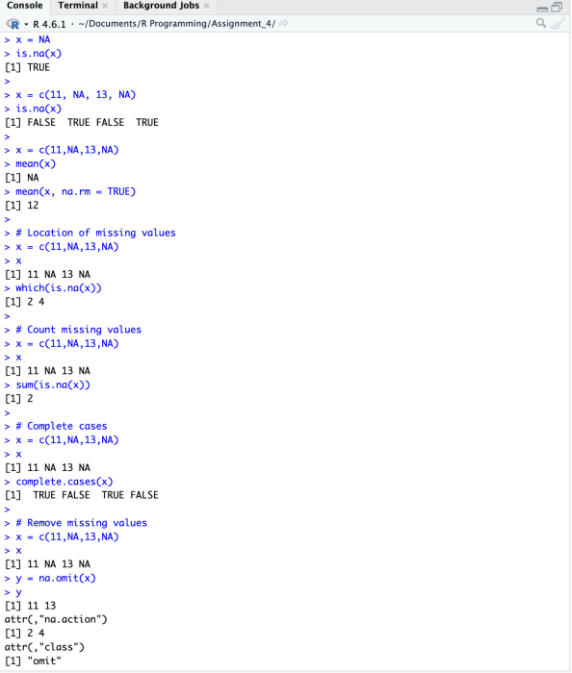
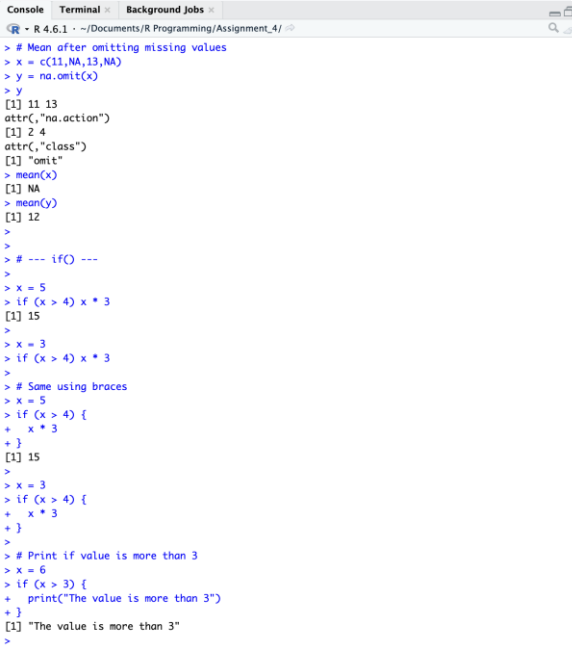


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NPTEL Week	5

Screenshots	 <pre> R 4.6.1 ~ /Documents/R Programming/Assignment_4/ > x = NA > is.na(x) [1] TRUE > > x = c(11, NA, 13, NA) > is.na(x) [1] FALSE TRUE FALSE TRUE > > x = c(11, NA, 13, NA) > mean(x) [1] NA > mean(x, na.rm = TRUE) [1] 12 > > # Location of missing values > x = c(11, NA, 13, NA) > x [1] 11 NA 13 NA > which(is.na(x)) [1] 2 4 > > # Count missing values > x = c(11, NA, 13, NA) > x [1] 11 NA 13 NA > sum(is.na(x)) [1] 2 > > # Complete cases > x = c(11, NA, 13, NA) > x [1] 11 NA 13 NA > complete.cases(x) [1] TRUE FALSE TRUE FALSE > > # Remove missing values > x = c(11, NA, 13, NA) > x [1] 11 NA 13 NA > y = na.omit(x) > y [1] 11 13 attr(,"na.action") [1] 2 4 attr(,"class") [1] "omit" </pre>
	 <pre> R 4.6.1 ~ /Documents/R Programming/Assignment_4/ > # Mean after omitting missing values > x = c(11, NA, 13, NA) > y = na.omit(x) > y [1] 11 13 attr(,"na.action") [1] 2 4 attr(,"class") [1] "omit" > mean(x) [1] NA > mean(y) [1] 12 > > > # --- if() --- > > x = 5 > if (x > 4) x * 3 [1] 15 > > x = 3 > if (x > 4) x * 3 > > # Same using braces > x = 5 > if (x > 4) { + x * 3 + } [1] 15 > > x = 3 > if (x > 4) { + x * 3 + } > > # Print if value is more than 3 > x = 6 > if (x > 3) { + print("The value is more than 3") + } [1] "The value is more than 3" > </pre>

```

Console Terminal Background Jobs
R - R 4.6.1 - ~/Documents/R Programming/Assignment_4/
> x = 2
> if (x > 3) {
+   print("The value is more than 3")
+ }
>
>
> # --- if else() ---
>
> x = 5
> if (x == 3) {
+   x = x - 1
+ } else {
+   x = 2*x
+ }
> x
[1] 10
>
> x = 3
> if (x == 3) {
+   x = x - 1
+ } else {
+   x = 2*x
+ }
> x
[1] 2
>
> x = 6
> if (x > 3) {
+   print("The value is more than 3")
+ } else {
+   print("The value is less than 3")
+ }
[1] "The value is more than 3"
>
> x = 2
> if (x > 3) {
+   print("The value is more than 3")
+ } else {
+   print("The value is less than 3")
+ }
[1] "The value is less than 3"

```

```

Console Terminal Background Jobs
R - R 4.6.1 - ~/Documents/R Programming/Assignment_4/
>
> # --- Nested if else if() ---
>
> x = 5
> if (x == 3) {
+   x = x - 1
+ } else if (x < 3) {
+   x = x + 5
+ } else {
+   x = 2*x
+ }
> x
[1] 10
>
> x = 2
> if (x == 3) {
+   x = x - 1
+ } else if (x < 3) {
+   x = x + 5
+ } else {
+   x = 2*x
+ }
> x
[1] 7
>
> x = 3
> if (x == 3) {
+   x = x - 1
+ } else if (x < 3) {
+   x = x + 5
+ } else {
+   x = 2*x
+ }
> x
[1] 2
>
>
> # --- ifelse() ---
>
> x = 1:10
> x
[1] 1 2 3 4 5 6 7 8 9 10
> ifelse(x < 6, x^2, x + 1)
[1] 1 4 9 16 25 7 8 9 10 11
>

```

ConsoleTerminalBackground Jobs

R · R 4.6.1 · ~\Documents\R Programming\Assignment_4\

> x = c(7,9,8,4)

> ifelse(x %% 2 == 0, "even number", "odd number")

[1] "odd number" "odd number" "even number" "even number"

>

> # --- switch() ---

>

> switch(2, "apple", "banana", "orange")

[1] "banana"

> switch(1, "apple", "banana", "orange")

[1] "apple"

>

> switch(

+ "colour",

+ "colour" = "blue",

+ "gender" = "male",

+ "volume" = 50

+)

[1] "blue"

>

> switch(

+ "volume",

+ "colour" = "blue",

+ "gender" = "male",

+ "volume" = 50

+)

[1] 50

>

> switch(4, "apple", "banana", "orange")

>

> switch(

+ "size",

+ "colour" = "blue",

+ "gender" = "male",

+ "volume" = 50

+)

>

> # --- which() ---

>

> x = c(10,15,8,14,6,12)

> x

[1] 10 15 8 14 6 12

> which(x == 14)

[1] 4

> which(x != 12)

[1] 1 2 3 4 5

> which(x > 10)

[1] 2 4 6

>

> x = matrix(nrow=3, ncol=3, data=1:9)

> x

[,1] [,2] [,3]

[1,] 1 4 7

[2,] 2 5 8

[3,] 3 6 9

> which.min(x)

[1] 1

> which.max(x)

[1] 9

>

> x = matrix(nrow=3, ncol=3, data=1:9)

> x

[,1] [,2] [,3]

[1,] 1 4 7

[2,] 2 5 8

[3,] 3 6 9

> which(x %% 2 == 1)

[1] 1 3 5 7 9

> which(x %% 2 == 1, arr.ind = TRUE)

row col

[1,] 1 1

[2,] 3 1

[3,] 2 2

[4,] 1 3

[5,] 3 3

> |

R · Global Environment

Data

dnum [1:3, 1:3] 5 0 0 0 5 0 0 0 5

warpbreaks54 obs. of 3 variables

xint [1:3, 1:3] 1 2 3 4 5 6 7 8 9

xtint [1:2, 1:4] 1 2 3 4 5 6 7 8

Values

x1num [1:4] 1 2 3 4

x2num [1:4] 1 4 9 16

ynum [1:2] 11 13

